

# FoC Motor Controller

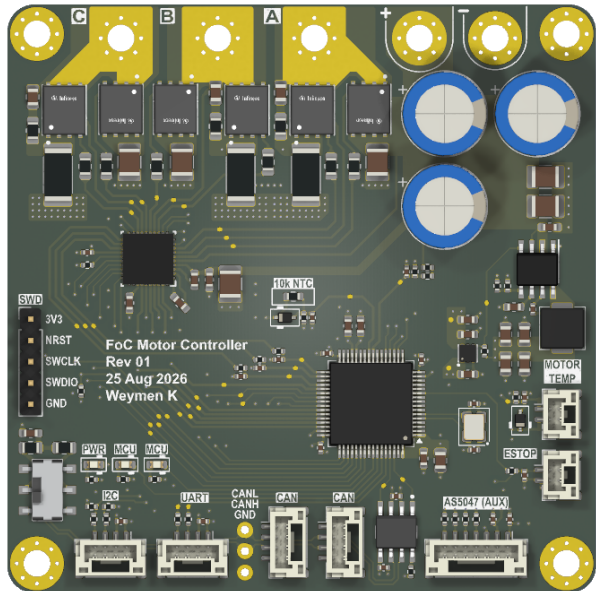
2026-08-28

Variant:

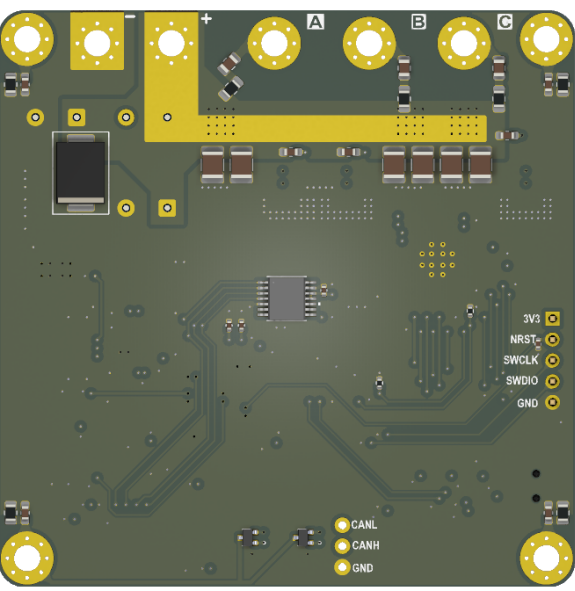
Rev 01

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| 10   | Sensing - Battery      | 20   |                           | 30   |       | 40   |       |

TOP VIEW



BOTTOM VIEW



## NOTES

Add a comment here

Not fitted components are marked as **X**

## DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

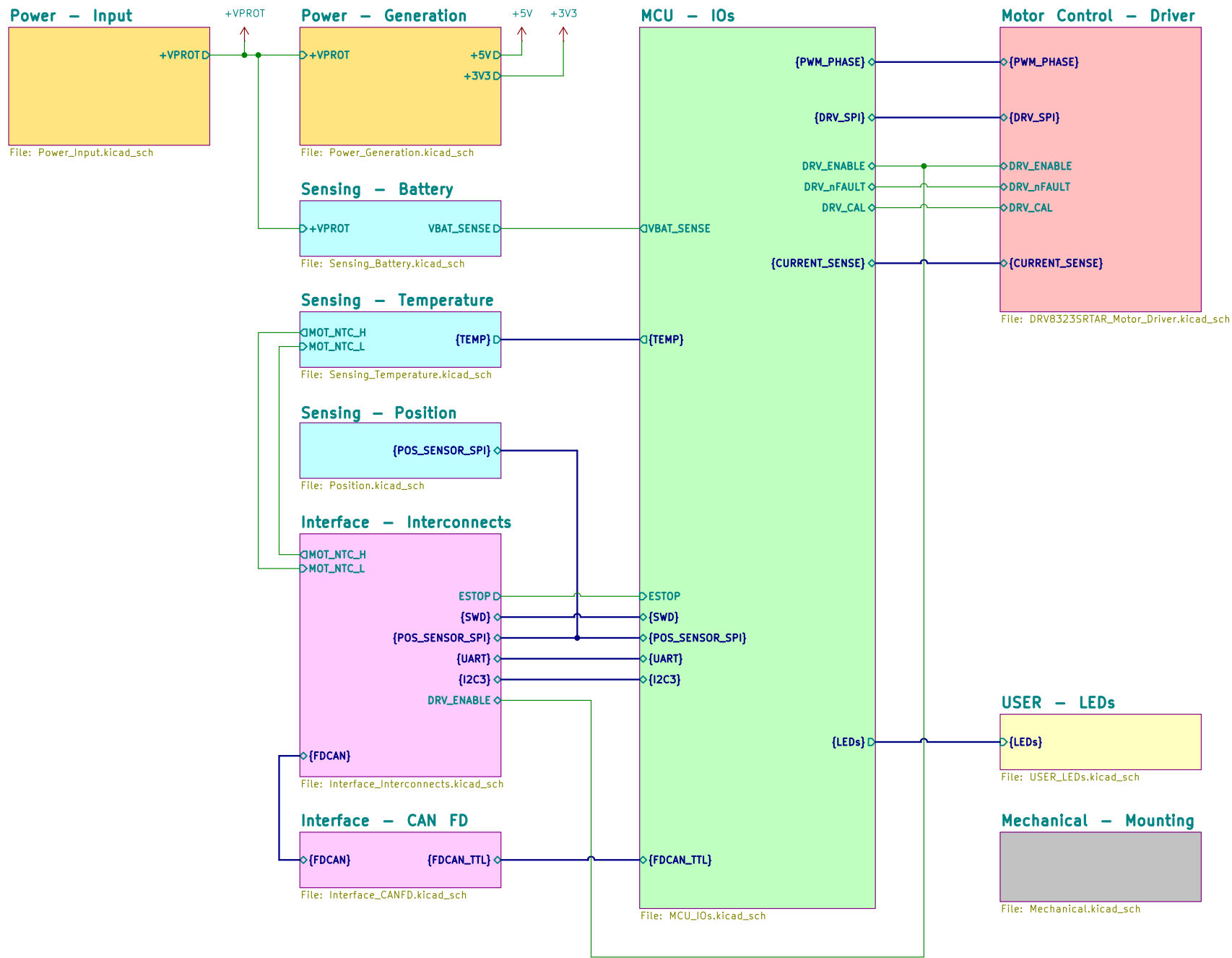
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

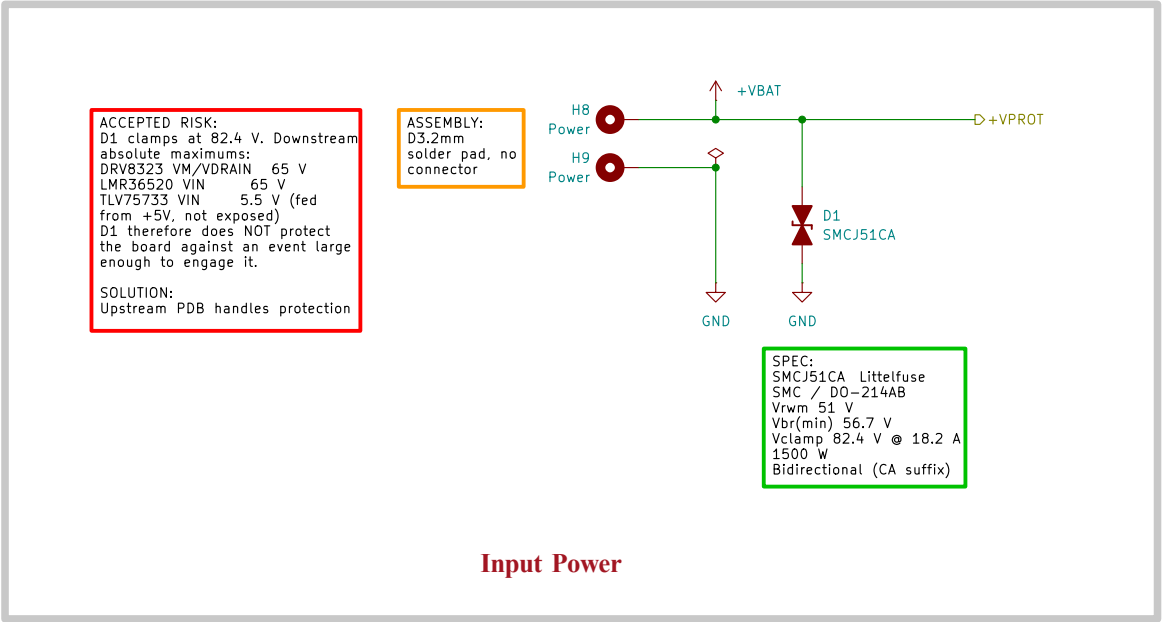
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|              |                                | -                    |                    |                      |  |
|              |                                | Board Name:          |                    | Project Name:        |  |
|              |                                | FoC Motor Controller |                    | FoC Motor Controller |  |
| Sheet Title: | File Name:                     | Designer:            | Date:              | Revision:            |  |
| Cover Page   | FoC Motor Controller.kicad_sch | Weymen Koo           | Last Modified Date | 01                   |  |
| Sheet Path:  | Reviewer:                      |                      | Size:              | Sheet:               |  |
| /            | N/A                            |                      | A3                 | 1 of 14              |  |

# [2] Project Architecture



|              |                        |  |                      |                                |                      |                    |
|--------------|------------------------|--|----------------------|--------------------------------|----------------------|--------------------|
|              | Comments:              |  | Company:             |                                | Variant:             |                    |
|              |                        |  | -                    |                                |                      |                    |
|              |                        |  | Board Name:          |                                | Project Name:        |                    |
|              |                        |  | FoC Motor Controller |                                | FoC Motor Controller |                    |
| Sheet Title: | Project Architecture   |  | File Name:           | Project Architecture.kicad_sch | Designer:            | Weymen Koo         |
|              |                        |  |                      |                                | Date:                | Last Modified Date |
| Sheet Path:  | /Project Architecture/ |  | Reviewer:            | N/A                            | Size:                | A3                 |
|              |                        |  |                      |                                | Sheet:               | 2 of 14            |

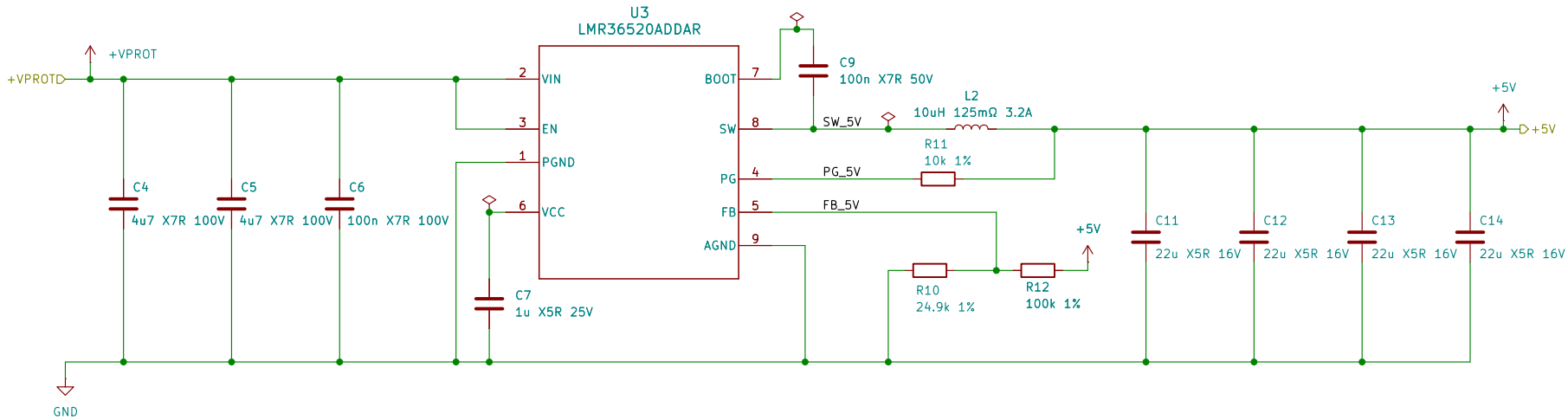
# [3] Power - Input



|  |                                                     |  |                                     |                         |                                       |                   |
|--|-----------------------------------------------------|--|-------------------------------------|-------------------------|---------------------------------------|-------------------|
|  | Comments:                                           |  | Company:                            |                         | Variant:                              |                   |
|  |                                                     |  | Board Name:<br>FoC Motor Controller |                         | Project Name:<br>FoC Motor Controller |                   |
|  | Sheet Title:                                        |  | File Name:<br>Power_Input.kicad_sch | Designer:<br>Weymen Koo | Date:                                 | Revision:         |
|  | Sheet Path:<br>/Project Architecture/Power - Input/ |  |                                     | Reviewer:<br>N/A        | Size:<br>A4                           | Sheet:<br>3 of 14 |

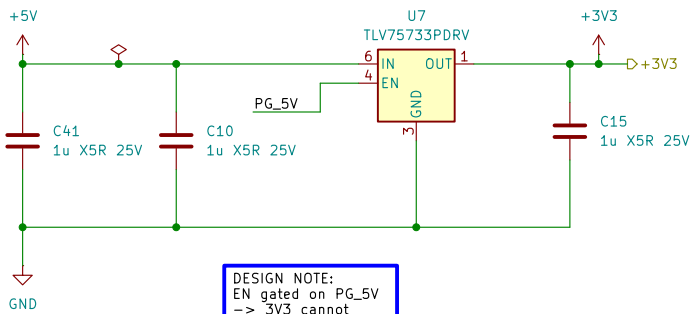
# [4] Power - Generation

SPEC:  
LMR36520ADDAR TI  
SO PowerPAD-8 (DDA)  
4.2-65 V in  
2 A  
f<sub>sw</sub> 400 kHz fixed  
Vref 1.0 V



DESIGN NOTE:  
5V BUCK  
 $V_{out} = 1.0 \times (1 + R_{12}/R_{10})$   
 $= 1.0 \times (1 + 100k/24.9k) = 5.016 V$   
 $dIL = V_{out}(V_{in}-V_{out})/(V_{in} \times L \times f_{sw})$   
 $= 5(50.4-5)/(50.4 \times 10u \times 400k) = 1.13 A \text{ pk-pk}$   
 $I_{pk} = I_{out} + 0.56 A$  L2 sat 3.2 A -> OK to 2.6 A load.  
Cin(eff): 2 x 4u7 X7R 1210 at 50 V DC bias -> 40% -> -3.8 uF  
Cout(eff): 4 x 22u X5R 0805 at 5 V DC bias -> 70% -> -62 uF

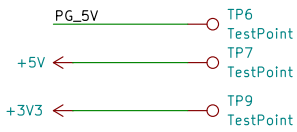
5V Buck Converter



DESIGN NOTE:  
3V3 BUDGET AND LDO THERMAL  
 $P_d = (5.02 - 3.30) \times I = 1.72 \times 500mA = 0.86 W$   
Theta-JA (WS0N-6, 2 oz, thermal pad vias)  
~> 78 C/W  
dT = 67 C, Tj at 25 C ambient = 92 C.  
Limit 125 C.  
Require external heatsink if operating in warm ambient

SPEC:  
TLV75733PDRV TI  
WS0N-6 2x2 (DRV)  
Vin 2.5-5.5 V  
1 A  
Vdo ~240 mV @ 1 A  
fixed 3.3 V  
EN gated on PG\_5V -> 3V3 cannot precede 5V.

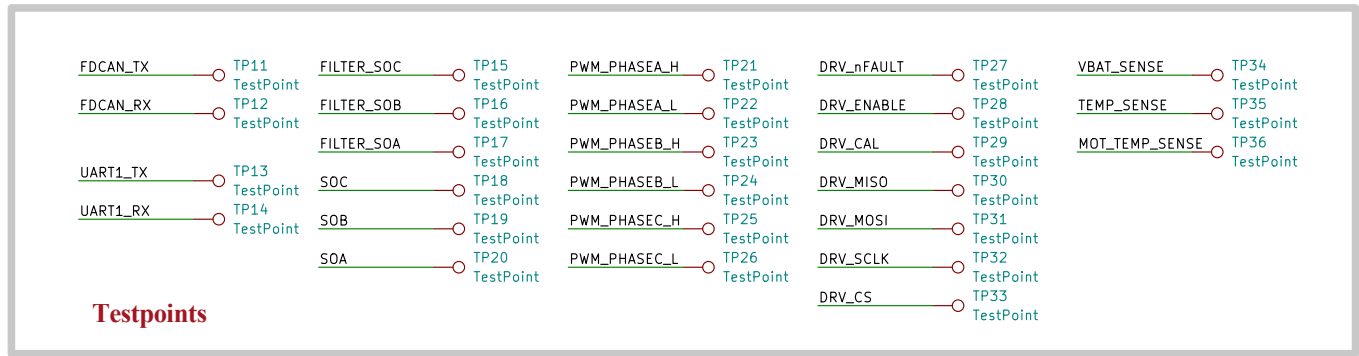
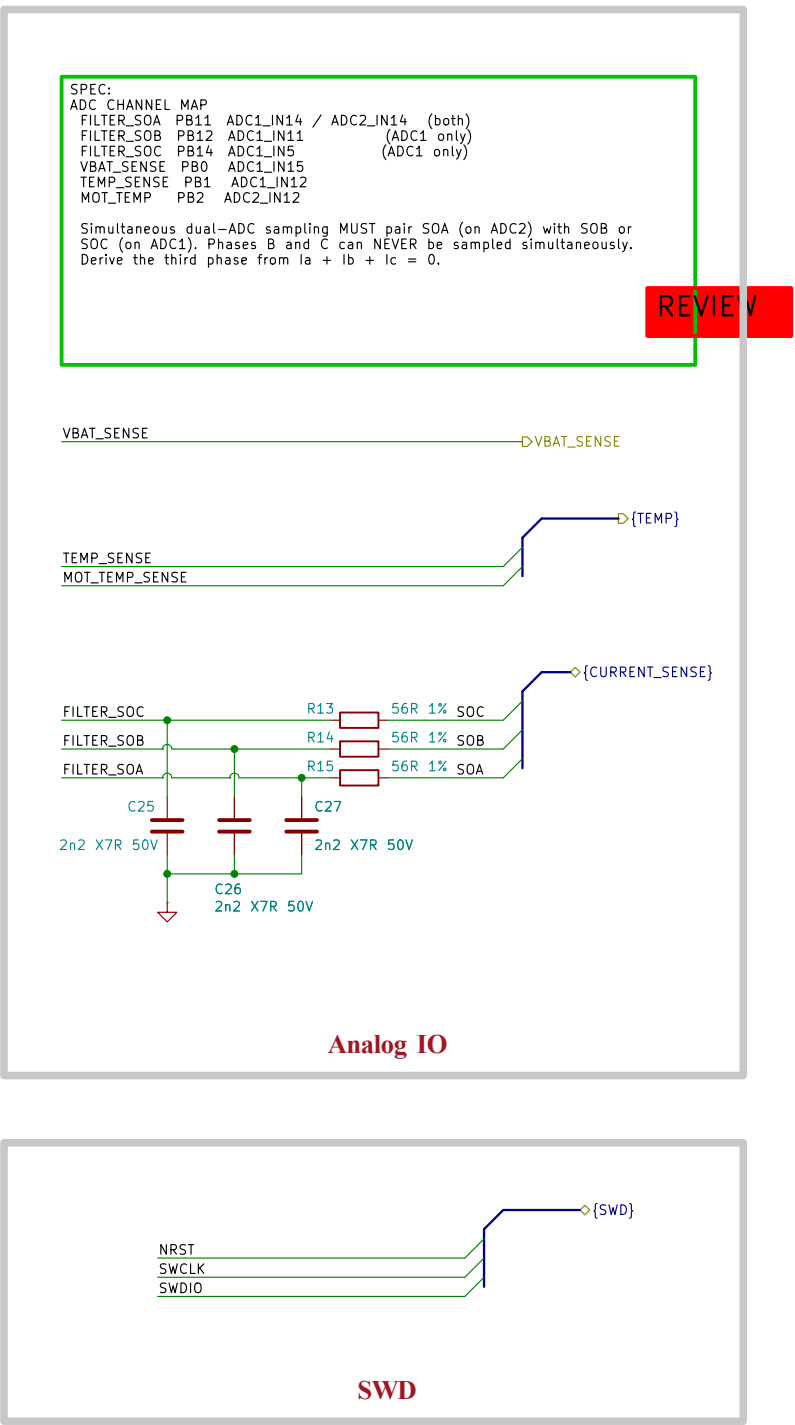
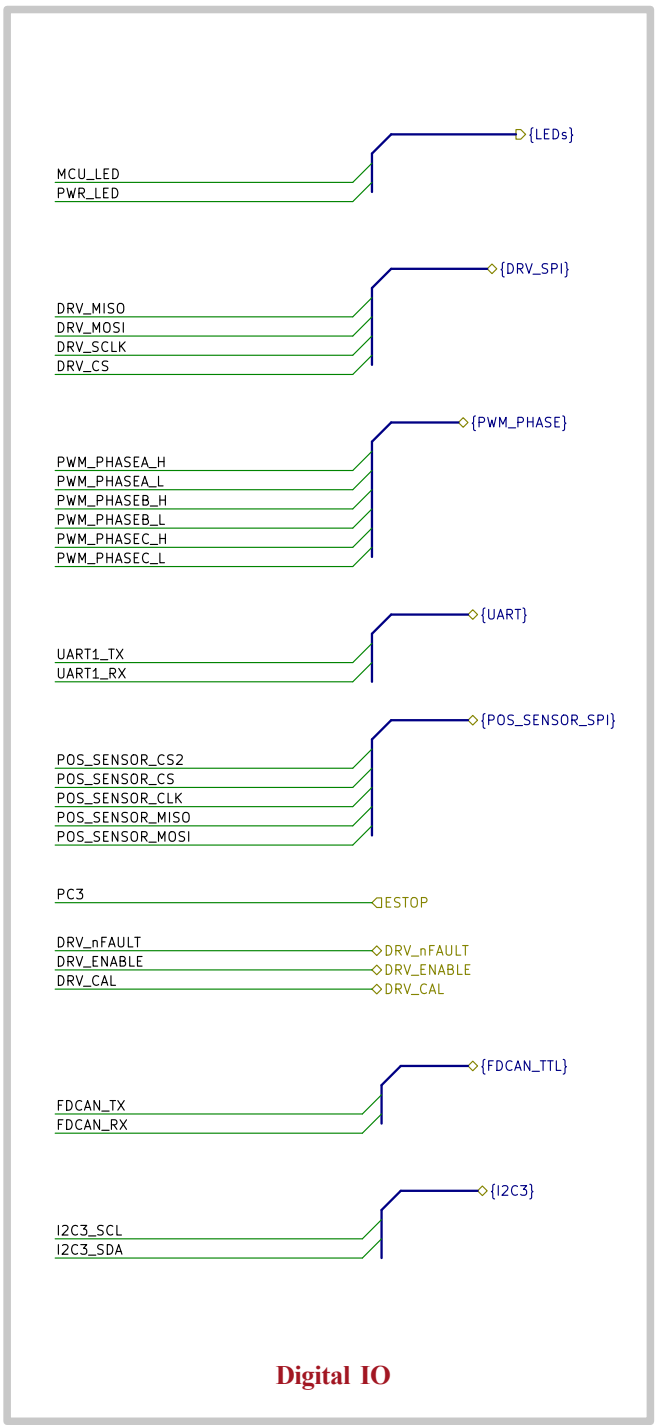
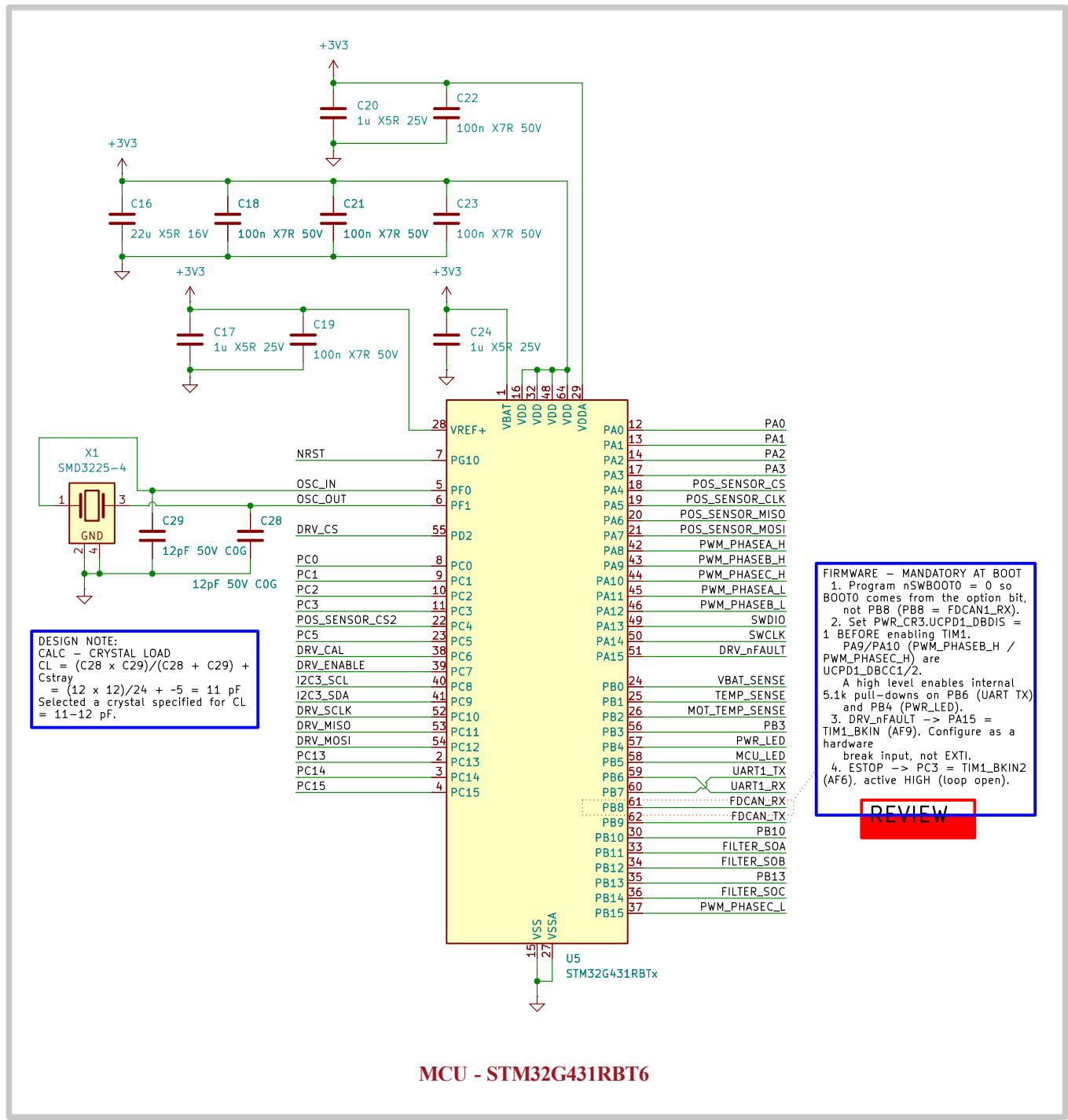
3V3 Linear Regulator



Testpoints

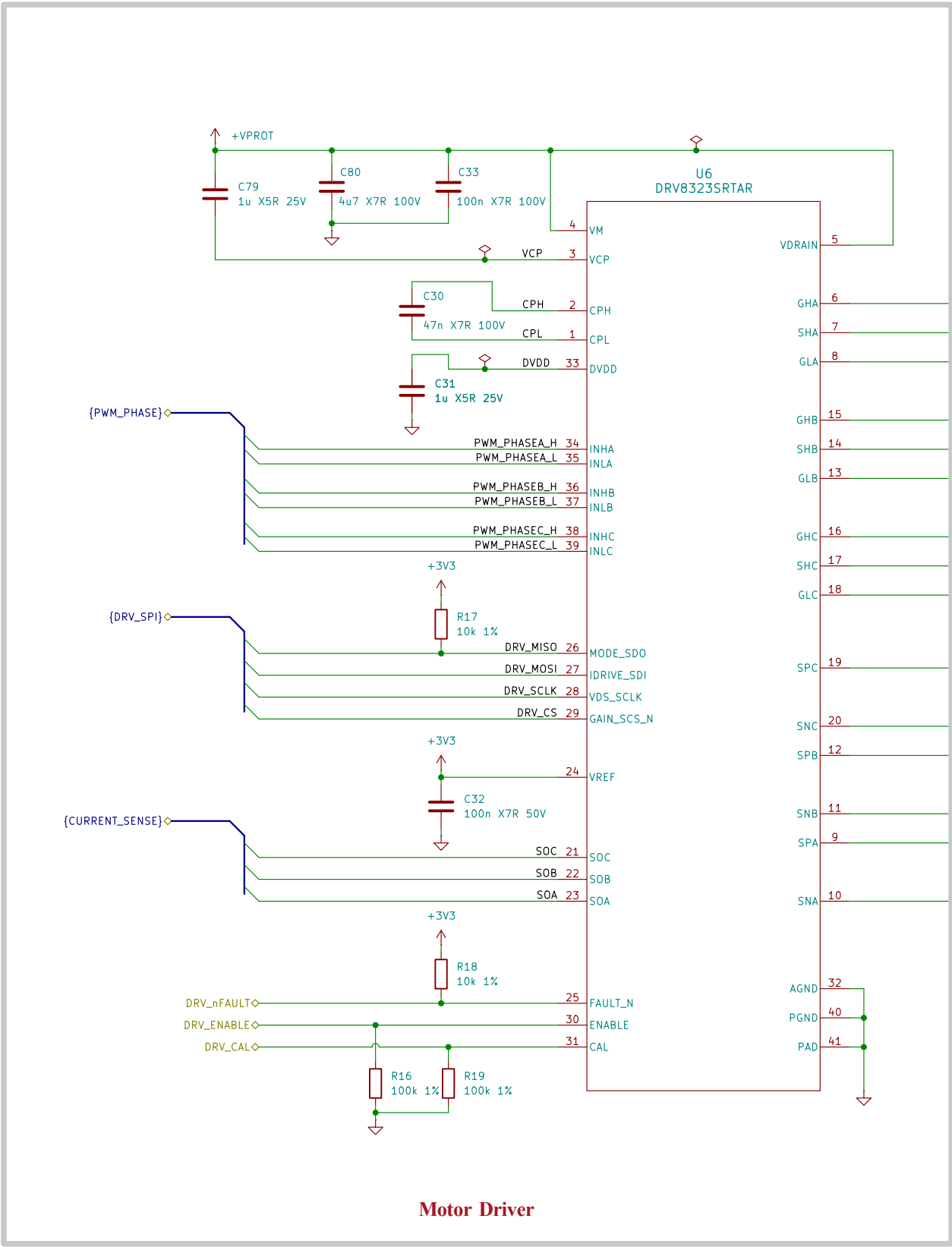
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|  |                                                              |  | Board Name:<br><b>FoC Motor Controller</b>   |                             | Project Name:<br><b>FoC Motor Controller</b> |                        |  |                                     |  |  |
|  | Sheet Title:                                                 |  | File Name:<br><br>Power_Generation.kicad_sch | Designer:<br><br>Weymen Koo |                                              | Date:                  |  | Revision:                           |  |  |
|  | Sheet Path:<br><br>/Project Architecture/Power - Generation/ |  |                                              | Reviewer:<br><br>N/A        |                                              | Size:<br><br><b>A3</b> |  | Sheet:<br><br><b>4</b> of <b>14</b> |  |  |

# [5] MCU - IOs

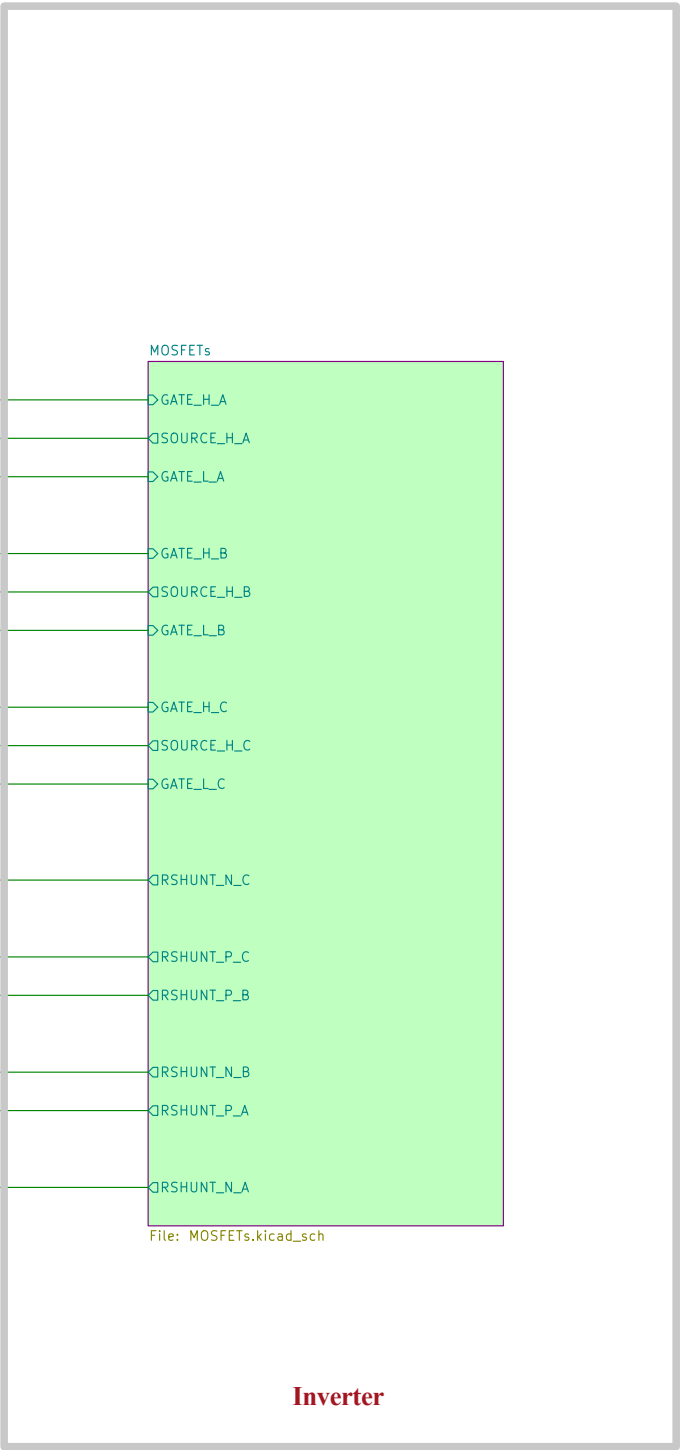


|                                                  |                                     |  |                                       |                   |           |  |
|--------------------------------------------------|-------------------------------------|--|---------------------------------------|-------------------|-----------|--|
| Sheet Title:<br>/Project Architecture/MCU - IOs/ | Comments:                           |  | Company:                              |                   | Variant:  |  |
|                                                  | Board Name:<br>FoC Motor Controller |  | Project Name:<br>FoC Motor Controller |                   | Revision: |  |
|                                                  | File Name:<br>MCU_IOs.kicad_sch     |  | Designer:<br>Weymen Koo               | Date:             | Revision: |  |
|                                                  | Reviewer:<br>N/A                    |  | Size:<br>A3                           | Sheet:<br>5 of 14 |           |  |

[6] Motor Driver - DRV8323SRTAR



Motor Driver



Inverter

SPEC:  
DRV8323SRTAR TI WQFN-40 6x6 (RTA)  
VM 6-60 V rec, 65 V abs max  
3x CSA  
SPI  
1 A/2 A gate drive  
Digital pin abs max 5.75 V (independent of VM)  
SPx/SNx: +/-1 V continuous, +/-3 V transient 200 ns

DESIGN NOTE:  
CALC - CHARGE PUMP  
 $I_{VCP} = 3 \times Q_g(HS) \times f_{sw} = 3 \times [FILL] \text{ nC} \times [FILL] \text{ kHz} = [FILL] \text{ mA}$   
DRV8323 supplies 25 mA. Margin = [FILL] x.

REVIEW

- GHA TP37 TestPoint
- SHA TP38 TestPoint
- GLA TP39 TestPoint
- GHB TP40 TestPoint
- SHB TP41 TestPoint
- GLB TP42 TestPoint
- GHC TP43 TestPoint
- SHC TP44 TestPoint
- GLC TP45 TestPoint

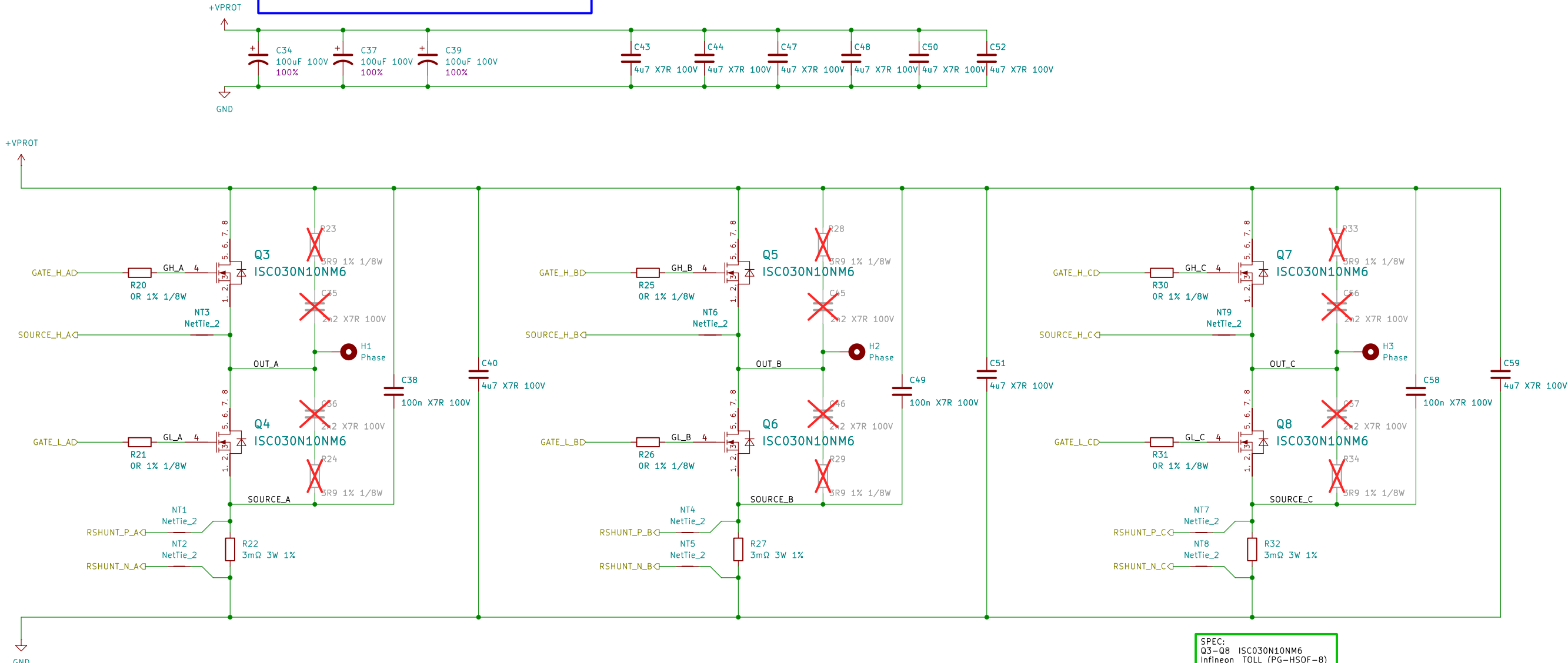
Test Points & Design Notes

|                                               |                      |           |                                     |            |          |           |
|-----------------------------------------------|----------------------|-----------|-------------------------------------|------------|----------|-----------|
|                                               | Comments:            |           | Company:                            |            | Variant: |           |
|                                               | Board Name:          |           | Project Name:                       |            |          |           |
|                                               | FoC Motor Controller |           | FoC Motor Controller                |            |          |           |
|                                               | Sheet Title:         |           | File Name:                          | Designer:  | Date:    | Revision: |
|                                               |                      |           | DRV8323SRTAR_Motor_Driver.kicad_sch | Weymen Koo |          |           |
| Sheet Path:                                   |                      | Reviewer: |                                     | Size:      | Sheet:   |           |
| /Project Architecture/Motor Control - Driver/ |                      | N/A       |                                     | A3         | 6 of 14  |           |

## [7] MOSFETs

DESIGN NOTE:  
 CALC = DC LINK  
 Bulk  $3 \times 100 \mu\text{F} = 300 \mu\text{F}$  (polymer, negligible DC derating)  
 MLCC  $9 \times 4\mu\text{F} = 42.3 \mu\text{F}$  nominal  
 at 50 V DC bias, 1210 X7R  $\sim 0.4 \rightarrow 16.9 \mu\text{F}$  effective  
 Bus ripple  $dV = I_{dc} / (f_{sw} \times C) = [\text{FILL}] / ([\text{FILL}] \times 316\mu) = [\text{FILL}]$   
 Ripple current requirement:  $[\text{FILL}] \text{ A RMS vs } 3 \times [\text{FILL}] \text{ A rated}$

## REVIEW



DESIGN NOTE:  
CALC - CURRENT SENSE  
Rshunt = 3 mR, bidirectional, VREF = 3.3 V, zero at 1.65 V  
Gain 5 -> +/-110 A      Gain 20 -> +/-27.5 A  
Gain 10 -> +/-55 A      Gain 40 -> +/-13.75 A  
SELECTED: gain 20 V/V  
ADC resolution at gain 20:  
(3.3/4096) / 20 / 3 mR = 13.4 mA per LSB  
Vsense at 20 A = 60 mV (within +/-1 V continuous spec)

REVIEW  
REVIEW

DESIGN NOTE:  
 CALC - CONDUCTION LOSS  
 $R_{ds(on)}$  at  $T_j$  125 °C =  $3.0 \text{ m}\Omega \times 1.5 = 4.5 \text{ m}\Omega$   
 $P_{cond} = 3 \times I_{ph\_rms}^2 \times R_{ds(on)} = 3 \times [FILL]^2 \times 4.5 \text{ mW}$   
 [FILL] W  
 Shunt  $P = I_{ph\_rms}^2 \times 3 \text{ m}\Omega = [FILL] \text{ W}$  each (3 W rated)  
 Vsense at 20 A = 60 mV

REVIEW

SPEC:  
Q3-Q8 ISC030N10NM6  
Infineon TOLL (PG-HSOF-8)  
100 V  
3.0 mR typ @ 10 V  
Qg [FILL] nC  
Qgd [FILL] nC

REVIEW

|                                                                      |                                            |                         |                                              |                   |
|----------------------------------------------------------------------|--------------------------------------------|-------------------------|----------------------------------------------|-------------------|
| Comments:                                                            | Company:                                   |                         | Variant:                                     |                   |
|                                                                      | Board Name:<br><b>FoC Motor Controller</b> |                         | Project Name:<br><b>FoC Motor Controller</b> |                   |
| Sheet Title:                                                         | File Name:<br>MOSFETs.kicad_sch            | Designer:<br>Weymen Koo | Date:                                        | Revision:         |
| Sheet Path:<br>/Project Architecture/Motor Control - Driver/MOSFETs/ |                                            | Reviewer:<br>N/A        | Size:<br>A3                                  | Sheet:<br>7 of 14 |

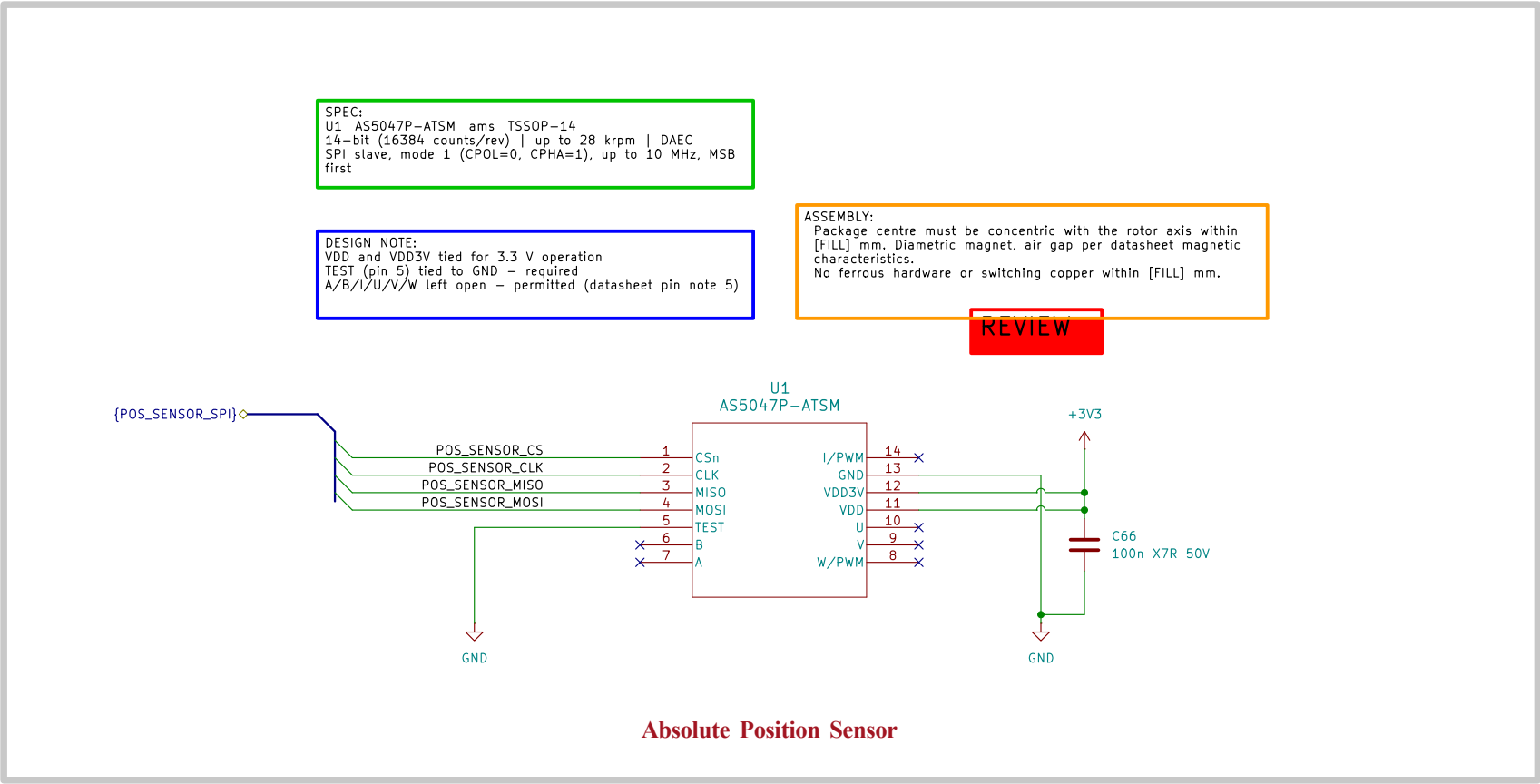
# [8] Sensing - Position

SPEC:  
U1 AS5047P-ATSM ams TSSOP-14  
14-bit (16384 counts/rev) | up to 28 krpm | DAEC  
SPI slave, mode 1 (CPOL=0, CPHA=1), up to 10 MHz, MSB first

DESIGN NOTE:  
VDD and VDD3V tied for 3.3 V operation  
TEST (pin 5) tied to GND - required  
A/B/I/U/V/W left open - permitted (datasheet pin note 5)

ASSEMBLY:  
Package centre must be concentric with the rotor axis within [FILL] mm. Diametric magnet, air gap per datasheet magnetic characteristics.  
No ferrous hardware or switching copper within [FILL] mm.

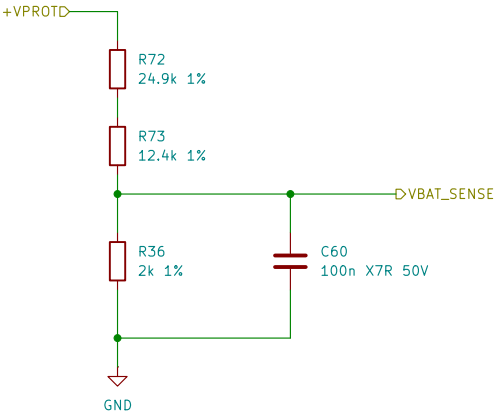
REVIEW



|  |                                                          |                                            |                         |                                              |                          |
|--|----------------------------------------------------------|--------------------------------------------|-------------------------|----------------------------------------------|--------------------------|
|  | Comments:                                                | Company:                                   |                         | Variant:                                     |                          |
|  |                                                          | Board Name:<br><b>FoC Motor Controller</b> |                         | Project Name:<br><b>FoC Motor Controller</b> |                          |
|  | Sheet Title:                                             | File Name:<br>Position.kicad_sch           | Designer:<br>Weymen Koo | Date:                                        | Revision:                |
|  | Sheet Path:<br>/Project Architecture/Sensing - Position/ | Reviewer:<br>N/A                           |                         | Size:<br><b>A4</b>                           | Sheet:<br><b>8 of 14</b> |

# [9] Sensing - Battery

DESIGN NOTE:  
Ratio =  $R36 / (R72 + R73 + R36) = 2 / (24.9 + 12.4 + 2) = 2 / 39.3 = 0.05089$   
at 50.4 V -> 2.565 V  
at 56.0 V -> 2.850 V  
clips at VREF+ = 3.3 V when Vbus = 64.8 V  
Divider current at 50.4 V = 1.283 mA (continuous pack drain)  
P(R72) = 41 mW, P(R73) = 20 mW  
Rth = (R72+R73) || R36 = 1.90 k  
tau = 190 us, fc = 838 Hz

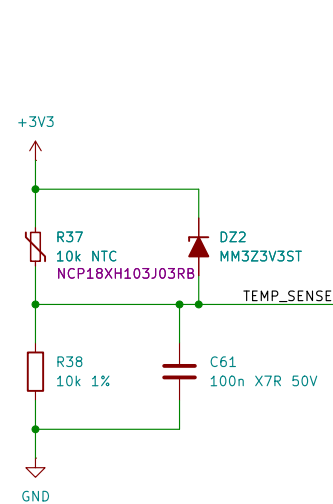


Battery Voltage Sensor

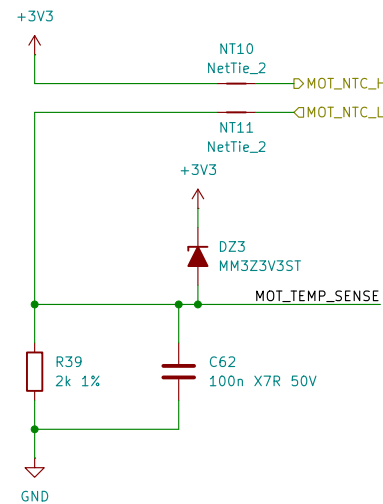
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|  |                                                         |  | Board Name:<br>FoC Motor Controller     |                         | Project Name:<br>FoC Motor Controller |                   |
|  | Sheet Title:                                            |  | File Name:<br>Sensing_Battery.kicad_sch | Designer:<br>Weymen Koo | Date:                                 | Revision:         |
|  | Sheet Path:<br>/Project Architecture/Sensing - Battery/ |  |                                         | Reviewer:<br>N/A        | Size:<br>A4                           | Sheet:<br>9 of 14 |

# [10] Sensing - Temperature

DESIGN NOTE:  
 CALC - 10k NTC ON-BOARD  
 NCP18XH103J03RB  
 $V = 3.3 \times R38 / (NTC + R38)$   
 -10 C (-42k) -> 0.63 V  
 25 C (10k) -> 1.65 V  
 100 C (-680R) -> 3.09 V  
 Self-heat: I = 165 uA, P = 0.27 mW,  
 <0.1 C rise. Negligible.



On-Board 10k NTC



Motor NTC

DESIGN NOTE:  
 CALC - MOTOR (external NTC on top, R39  
 2k to GND)  
 $V = 3.3 \times 2k / (NTC + 2k)$   
 25 C (10k) -> 0.55 V  
 100 C (-680R) -> 2.46 V



Comments:

Company:

Variant:

Board Name:

**FoC Motor Controller**

Project Name:

**FoC Motor Controller**

Sheet Title:

File Name:

Sensing\_Temperature.kicad\_sch

Designer:

Weymen Koo

Date:

Revision:

Sheet Path:

/Project Architecture/Sensing - Temperature/

Reviewer:

N/A

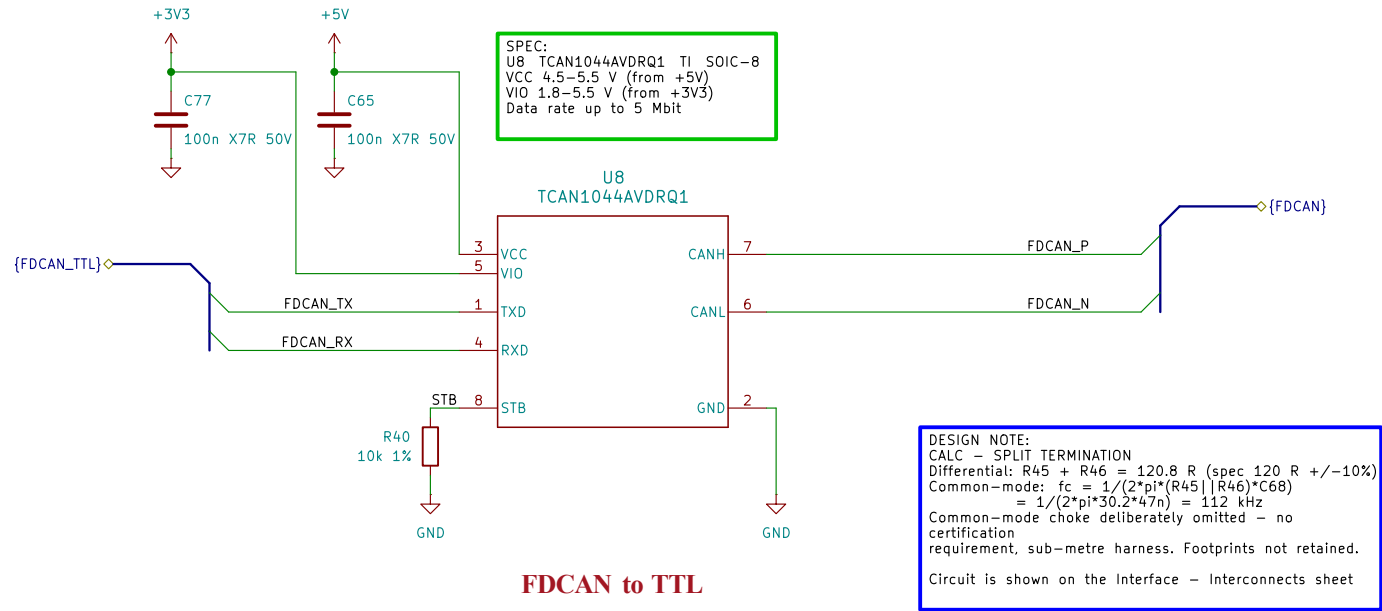
Size:

**A4**

Sheet:

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# [11] Interface - CAN FD



|  |                                                          |                                            |                         |                                              |           |
|--|----------------------------------------------------------|--------------------------------------------|-------------------------|----------------------------------------------|-----------|
|  | Comments:                                                | Company:                                   |                         | Variant:                                     |           |
|  |                                                          | Board Name:<br><b>FoC Motor Controller</b> |                         | Project Name:<br><b>FoC Motor Controller</b> |           |
|  | Sheet Title:                                             | File Name:<br>Interface_CANFD.kicad_sch    | Designer:<br>Weymen Koo | Date:                                        | Revision: |
|  | Sheet Path:<br>/Project Architecture/Interface - CAN FD/ | Reviewer:<br>N/A                           | Size:<br><b>A4</b>      | Sheet:<br><b>11 of 14</b>                    |           |

# [12] Interface - Interconnects

## DESIGN NOTE:

No ESD protection for prototype.  
Expected to be in lab environment.

## DESIGN NOTE:

Loop closed = PC3 low = run;  
Loop open OR cable unplugged  
= PC3 high = hardware break.

## SWD (2.54mm Pin Header)

## Temperature

## AUX2 Connector (Position Sensor SPI)

## FD\_CAN Connectors and Termination

## DESIGN NOTE:

UART PIN NAMING  
Net UART1\_TX lands on PB7 =  
USART1\_RX.  
Net UART1\_RX lands on PB6 =  
USART1\_TX.

## DESIGN NOTE:

J6 shares SPI1 (SCK/MISO/MOSI) with the  
on-board AS5047P (U1).  
Isolation is by chip select only: U1 on PA4,  
remote on PC4.  
AS5047P tri-states MISO when CSn is high  
(datasheet t\_02), so two  
slaves on one bus is valid - but firmware  
MUST NOT assert both  
chip selects.

## Comments:

## Company:

## Variant:

## Board Name:

FoC Motor Controller

## Project Name:

FoC Motor Controller

## Sheet Title:

## File Name:

Interface\_Interconnects.kicad\_sch

## Designer:

Weymen Koo

## Date:

## Revision:

## Sheet Path:

/Project Architecture/Interface - Interconnects/

## Reviewer:

N/A

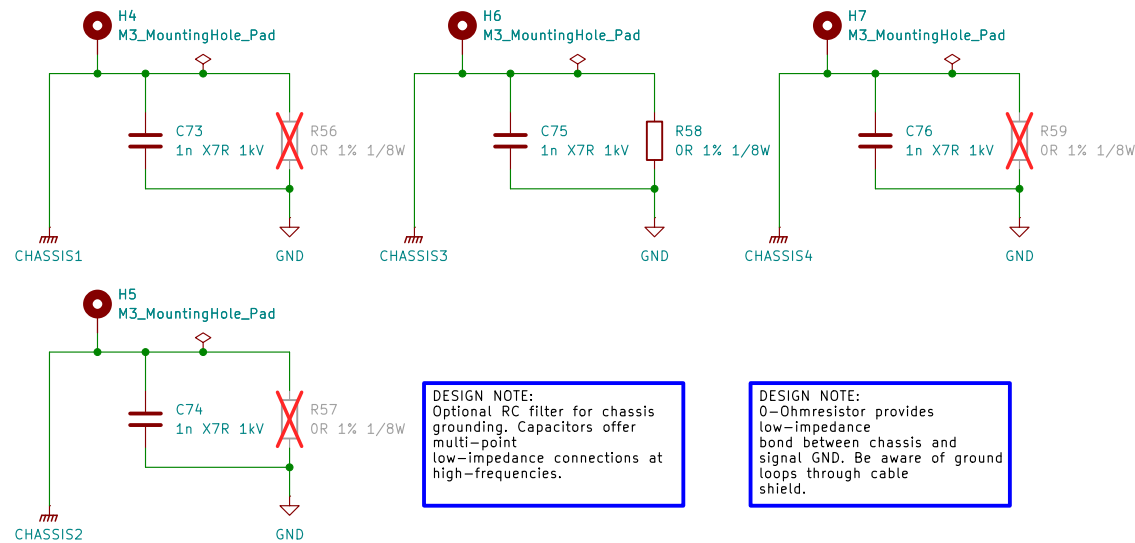
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A3

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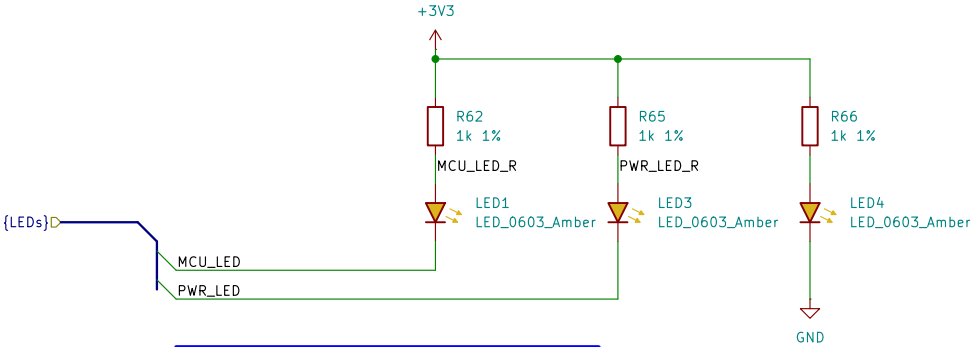
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# [13] Mechanical - Mounting Holes



|  |                                                             |                                            |                         |                                              |           |
|--|-------------------------------------------------------------|--------------------------------------------|-------------------------|----------------------------------------------|-----------|
|  | Comments:                                                   | Company:                                   |                         | Variant:                                     |           |
|  |                                                             | Board Name:<br><b>FoC Motor Controller</b> |                         | Project Name:<br><b>FoC Motor Controller</b> |           |
|  | Sheet Title:                                                | File Name:<br>Mechanical.kicad_sch         | Designer:<br>Weymen Koo | Date:                                        | Revision: |
|  | Sheet Path:<br>/Project Architecture/Mechanical - Mounting/ | Reviewer:<br>N/A                           | Size:<br><b>A4</b>      | Sheet:<br><b>13</b> of <b>14</b>             |           |

# [14] User - LEDs



DESIGN NOTE:  
LED1 MCU\_LED PB5, MCU-driven (sink)  
LED3 PWR\_LED PB4, MCU-driven (sink)  
LED4 hard-wired +3V3 -> R66 -> LED4 -> GND  
 $I = (3.3 - 2.0)/1k = 1.3 \text{ mA}$  each

|  |                                                   |                                     |                         |                                       |                    |
|--|---------------------------------------------------|-------------------------------------|-------------------------|---------------------------------------|--------------------|
|  | Comments:                                         | Company:                            |                         | Variant:                              |                    |
|  |                                                   | Board Name:<br>FoC Motor Controller |                         | Project Name:<br>FoC Motor Controller |                    |
|  | Sheet Title:                                      | File Name:<br>USER_LEDs.kicad_sch   | Designer:<br>Weymen Koo | Date:                                 | Revision:          |
|  | Sheet Path:<br>/Project Architecture/USER - LEDs/ |                                     | Reviewer:<br>N/A        | Size:<br>A4                           | Sheet:<br>14 of 14 |